UCT41042 – Practical for Introduction to Smart System

Department of Information & Communication Technology

Faculty of Technology

South Eastern University of Sri Lanka

Lab Sheet No: 01

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Title: Getting Started with ESP32 – Digital Input/Output and Wi-Fi Connectivity Basics

Aims: Introducing the fundamentals of ESP32 programming by developing and simulating digital input/output and Wi-Fi connectivity applications, including establishing Wi-Fi connections using both hardcoded credentials and the WiFiManager library, while using the Serial Monitor for debugging and real-time observation.

Tasks:

- Create and simulate ESP32 projects in the Wokwi simulator.
- Use the Serial Monitor for debugging and real-time observation.
- Connect the ESP32 to Wi-Fi using hardcoded SSID and password.
- Implement dynamic Wi-Fi configuration using the WiFiManager library.
- Combine Wi-Fi status with LED behavior for visual feedback and print connection details (IP address) only after successful connection.

<u>ESP32</u>

- ❖ Total Pins – 38 pins (on common 38-pin DevKit variants)
  - ➢ Ground Pins – 3 pins (GND)
  - ➢ Power Pins – 2 pins
    - ▪ 3V3 (3.3V output)
    - ▪ 5V / VIN (input voltage, usually 5V)
  - ➢ Other Pins
    - ▪ GPIO (General Purpose Input/Output) Pins
    - ▪ EN (Enable / Reset)
    - ▪ TX / RX (UART for serial communication)
    - ▪ Analog Input Pins (ADC)

Exercise 1: Connecting ESP32 to Wi-Fi

Part A: Hardcoded Wi-Fi Connection (Simple Method)

1. Create a new ESP32 project in Wokwi.
2. Write the code for hardcoded Wi-Fi connection using Wokwi-GUEST (open network, no password).
3. Run the simulation and observe the Serial Monitor (connection progress dots followed by successful connection and IP address).
4. Change the SSID to an incorrect name (e.g., "MyHomeWiFi") and observe the connection failure.

Part B: Using WiFiManager (Recommended for Real Projects)

1. Add the WiFiManager library in Wokwi (via Library Manager).
2. Write and run the WiFiManager code.
3. Observe the Serial Monitor for portal creation and connection status.
4. (For real hardware) Connect your phone/laptop to the “ESP32-Config-Portal” Access Point, open a browser at 192.168.4.1, and enter your Wi-Fi SSID and password.

Part C: Combine Wi-Fi status with LED behavior for visual feedback.

1. Blink the onboard LED slowly when not connected to Wi-Fi.
2. Blink the LED fast or keep it ON when successfully connected.
3. Print the IP address only after a successful connection.